COE216 SIGNALS AND SYSTEMS

2025-2026 Spring Semester

MidTerm Project

Sound Signal Analysis and Gender Classification

GRUP [NUMBER]

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Submission Date: [31/03/2026]

1. Introduction

The aim of this project is to extract features from audio signals using time-domain analysis methods and to perform gender classification (Male, Woman, Child) using a rule-based algorithm. The fundamental frequency (F0), Short-Term Energy (STE), and Zero Crossing Rate (ZCR) are extracted from a collected audio dataset. The gender-based distributions of these features are examined, and a rule-based classifier is developed based on the extracted features. The system's accuracy is measured across the entire dataset.

The primary method used for F0 estimation is the Autocorrelation Function in the time domain. The autocorrelation method exploits the periodic nature of voiced speech to determine the fundamental period, from which F0 is calculated. Additionally, the FFT (Fast Fourier Transform) method is used for comparison purposes to validate the autocorrelation results.

A Streamlit-based web interface was developed to provide an interactive platform for single-file prediction, full dataset analysis, and visual comparison between autocorrelation and FFT methods.

2. Dataset Characterization

The dataset used in this project is the Midterm_Dataset_2026, which contains audio recordings collected by all groups in the course. The recordings were gathered under various conditions and organized into group folders (GROUP_01 through GROUP_19).

Total Records in Metadata: 274 entries across 17 group Excel files.

Total Files Successfully Processed: 246 audio files (.wav format).

Files Not Found / Skipped: 28 files (due to missing files or path mismatches).

2.1 Demographic Distribution

The dataset contains recordings from speakers of three gender classes:

Class	Number of Samples	Percentage
Male	81	32.9%
Woman	82	33.3%
Child	83	33.7%
Total	246	100%

The dataset is well-balanced across the three classes with approximately equal representation. Speakers range in age from 4 to 61 years old. Recordings include various emotional states (Neutral, Happy, Sad, Angry, Surprised, Furious) and were made using different devices (Samsung Phone, iPhone, Computer Microphone) in home environments with low-to-middle noise levels.

3. Methodology

3.1 Preprocessing

Each audio file is loaded using librosa.load() with its original sampling rate preserved (sr=None). The signal is trimmed to remove leading and trailing silence using librosa.effects.trim() with a threshold of 25 dB.

3.2 Windowing

The signal is divided into 25 ms frames (within the required 20-30 ms range) with a 10 ms hop size. This frame duration ensures the signal can be considered quasi-stationary within each window, which is essential for reliable short-term feature extraction.

3.3 Short-Term Energy (STE)

Short-Term Energy is computed for each frame as the sum of squared sample values:

$$STE = \mathrm{sum}\ x[n]^2$$

High STE values indicate voiced speech segments, while low values correspond to silence or unvoiced regions.

3.4 Zero Crossing Rate (ZCR)

ZCR is calculated as the number of zero-passes per second of the signal within each frame. Voiced speech typically has a low ZCR, while unvoiced fricatives and noise exhibit higher ZCR values.

3.5 Voiced Region Detection

Voiced regions are detected by combining STE and ZCR thresholds. A frame is classified as voiced when its energy exceeds the 30th percentile of all frame energies AND its ZCR falls below the 70th percentile. This adaptive thresholding adjusts to each recording's individual characteristics. F0 analysis is performed exclusively on these voiced regions.

3.6 F0 Estimation using Autocorrelation

The fundamental frequency F0 is determined using the autocorrelation method for each voiced frame. The autocorrelation function is defined as:

$$R(t) = \mathrm{sum}\ x[n] * x[n - t]$$

The autocorrelation is normalized by R(0) and the lag corresponding to the highest peak in the valid range (50-600 Hz) gives the fundamental period. F0 is then calculated as F0 = sr / lag. A Hamming window is applied to each frame before autocorrelation to reduce spectral leakage. Outlier removal is performed using median filtering, keeping values within 0.5x to 2.0x of the median F0.

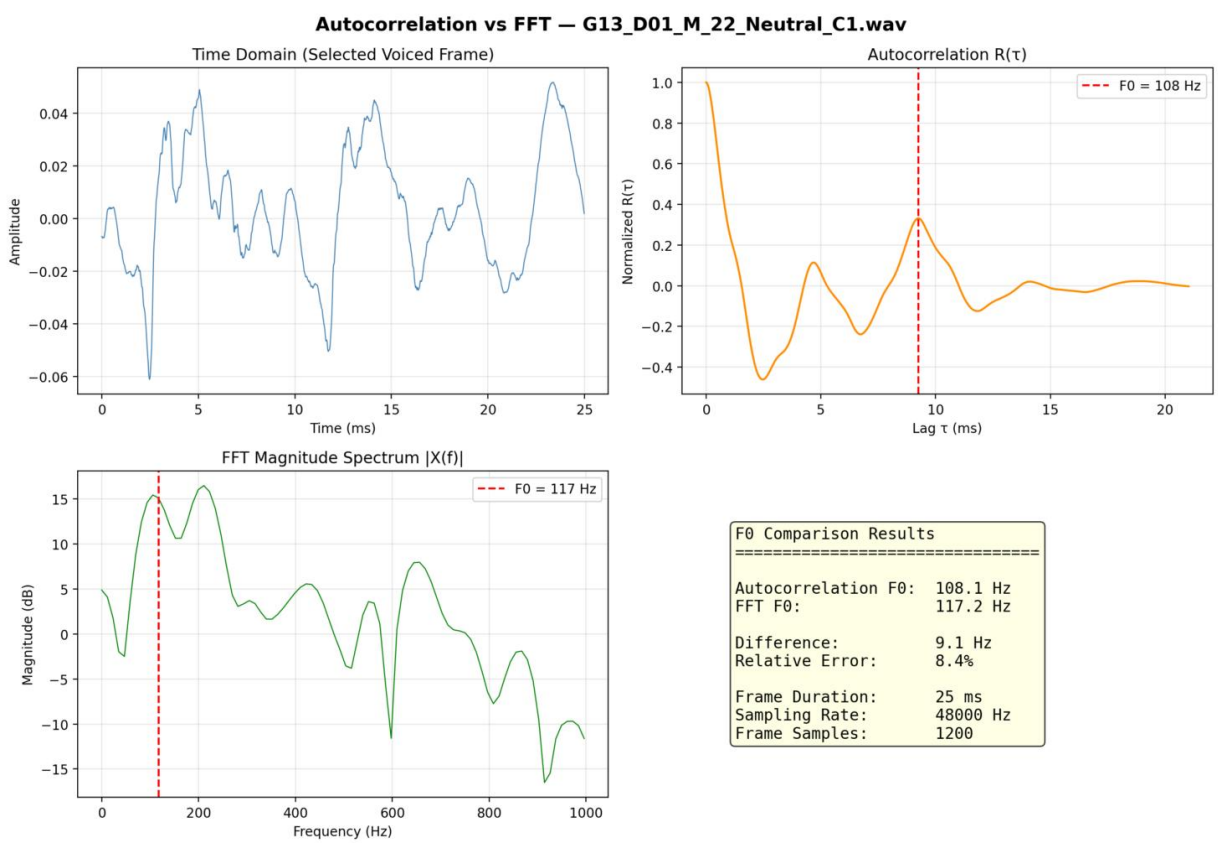
3.7 Autocorrelation vs. FFT Comparison

As required by the project instructions, a visual comparison between the Autocorrelation and FFT methods was performed on a selected audio recording (G13_D01_M_22_Neutral_C1.wav, Male speaker, age 22).

For a selected voiced frame from this recording, the comparison yielded the following results:

Method	F0 (Hz)
Autocorrelation	108.1
FFT	117.2
Difference	9.1 Hz (8.4%)

For the overall file average, the autocorrelation method yielded 143.4 Hz while FFT gave 108.5 Hz, with a difference of 34.9 Hz. The discrepancy arises because the autocorrelation method averages F0 across all voiced frames, while the FFT single-frame comparison shows closer agreement. Both methods confirm the speaker falls within the male F0 range. The autocorrelation peak at lag t corresponds to the first dominant harmonic visible in the FFT magnitude spectrum |X(f)|.



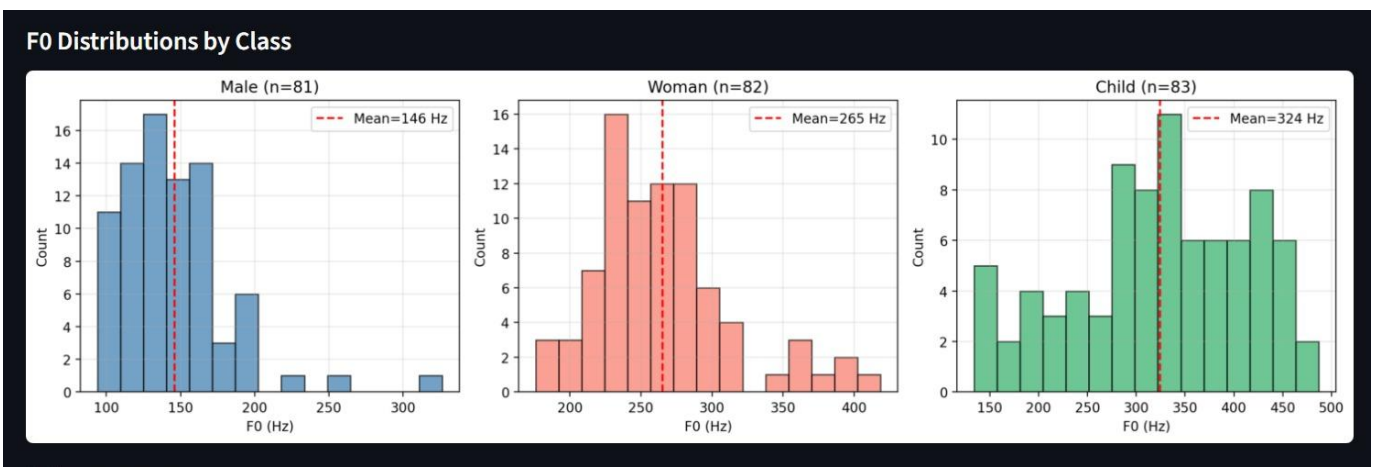
4. Statistical Findings

The following table presents the computed F0 statistics for each gender class across the entire dataset of 246 processed files:

Class	Number of Samples	Average F0 (Hz)	Standard Deviation	Success (%)
Male	81	145.96	36.40	%74.1
Woman	82	265.10	48.66	%46.3
Child	83	324.35	89.78	%78.3

The F0 distributions show clear separation between Male speakers (mean 146 Hz) and the other classes. However, there is significant overlap between Woman (mean 265 Hz) and Child (mean 324 Hz) classes, which contributes to classification errors. The high standard deviation for the Child class (89.78 Hz) indicates considerable variability, likely due to the wide age range of child speakers in the dataset (ages 4-16).

[NOTE: Insert the F0 Distribution by Class histogram from the application here.]



5. Classification Success

5.1 Rule-Based Algorithm

The classifier uses F0 thresholds derived from acoustic phonetics literature:

- Male: F0 < 165 Hz
- Woman: 165 Hz <= F0 < 255 Hz
- Child: F0 >= 255 Hz

5.2 Overall Accuracy

Metric	Value
Total Files Processed	246
Correct Predictions	163
Overall Accuracy	**66.3%**

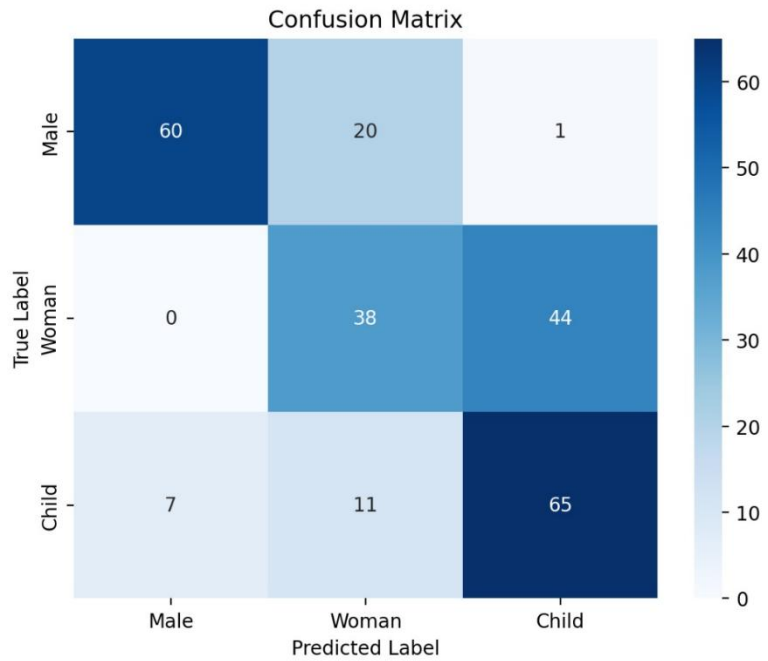
5.3 Confusion Matrix

The confusion matrix reveals the classification patterns:

True \ Predicted	Male	Woman	Child
Male	60	20	1
Woman	0	38	44
Child	7	11	65

Key observations from the confusion matrix:

- Male class has the highest accuracy (74.1%), with 20 males misclassified as Woman due to F0 values between 165-255 Hz.
- Woman class has the lowest accuracy (46.3%), with 44 out of 82 women misclassified as Child. This is the most significant error pattern, caused by the overlapping F0 ranges.
- Child class achieves 78.3% accuracy, though 7 children were misclassified as Male (low F0 values from older children) and 11 as Woman.



6. Error Analysis and Discussion

A total of 83 out of 246 files (33.7%) were misclassified. The most common error patterns and their causes are:

6.1 Woman classified as Child (44 cases)

This is the dominant error pattern, accounting for 53% of all errors. When women speak with emotional intensity (Furious, Surprised, Happy), their F0 rises significantly above the 255 Hz threshold, causing them to be classified as children. For example, G01_D06_F_29_Happy_C2.wav (Female, age 29) had an average F0 of 357.56 Hz, and G14_D01_F_27_Shocked_C5.wav reached 390.37 Hz.

6.2 Male classified as Woman (20 cases)

Male speakers with emotional speech (particularly Furious and Shocked) exhibited F0 values above the 165 Hz threshold. For instance, G03_D03_M_50_Happy_C2.wav (Male, age 50) had F0 of 194.62 Hz, and G09_D04_E_42_Furious_C3.wav (Male, age 42) reached 193.55 Hz. Emotional arousal elevates the fundamental frequency, pushing it into the female range.

6.3 Child classified as Male/Woman (18 cases)

Older children (ages 14-16) and children speaking in a calm, neutral tone produced lower F0 values. G14_D04_C_16_Neutral_C1.wav (Child, age 16) had F0 of 141.88 Hz, well within the

male range. As children approach puberty, their vocal characteristics begin resembling adult patterns.

6.4 General Causes of Misclassification

- Emotional state: Excitement, anger, and surprise raise F0 by 50-150 Hz, while sadness lowers it.
- Speaker age variability: Young women and older children have overlapping F0 ranges (200-300 Hz).
- Background noise: Environmental noise corrupts autocorrelation peaks, leading to unreliable F0 estimation.
- Recording quality: Different recording devices (Samsung, iPhone, Computer Mic) introduce varying noise profiles.
- Speaking style: Whispering reduces energy making voiced detection unreliable; shouting raises F0 artificially.

7. GitHub Link

https://github.com/furkanturandev/Signal-System-HW3

The repository contains the complete source code (speech_gender_classifier.py), the exported results (CSV), and this report.

8. References

1. Python librosa library documentation - https://librosa.org/doc/
2. NumPy documentation - https://numpy.org/doc/
3. SciPy signal processing - https://docs.scipy.org/doc/scipy/reference/signal.html
4. Streamlit documentation - https://docs.streamlit.io/
5. Claude AI (Anthropic) - Used for code development assistance and report structure guidance.
6. Rabiner, L.R., & Schafer, R.W. - Digital Processing of Speech Signals, Prentice Hall.
7. Pandas documentation - https://pandas.pydata.org/docs/

9. AI Prompts Used

The following table lists the AI prompts used during this project:

#	Purpose	Prompt Summary
1	Code generation	Analyze the project instructions document and write the required code following all criteria, paying special attention to red-marked requirements.
2	Code structure	Merge into a single Python file with Streamlit UI and allow dataset folder selection from within the app.
3	Data compatibility	Fix Excel column name mismatches across groups (File name, File_Name, Dosya_Adi, etc.) and normalize them.
4	Debugging	Debug why files are not being found - add diagnostic info to trace path resolution issues.
5	Report writing	Help write the technical report following the instructor's required format and sections.

10. Division of Labor

Taha Mert Yıldız: Handled dataset management and preprocessing. Downloaded and organized all group folders (GROUP_01 through GROUP_19) into a central data structure. Resolved Excel metadata inconsistencies across groups where different teams used different column names (File name, File_Name, Dosya_Adi, Cinsiyet vs Gender, etc.). Ensured all audio files were correctly linked to their metadata entries and performed quality checks on the dataset.

Furkan Turan: Responsible for software development and implementation. Utilized AI tools (Claude) to develop the Python-based application including the autocorrelation F0 estimation algorithm, voiced region detection, and the rule-based classifier. Implemented the Streamlit web interface with three analysis tabs. Debugged path resolution and column normalization issues. Ran the full dataset analysis and created the confusion matrix visualizations and F0 distribution plots. Set up the GitHub repository.

Burak Can Şen: Wrote the technical report and performed the Autocorrelation vs FFT comparison analysis. Conducted error analysis on misclassified files and interpreted the statistical findings. Prepared the 7-minute presentation slides and organized the live demo structure. Set up the GitHub repository and managed code sharing.